

HTML

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1. Introduction

To write some text...

- Text editors (Notepad, ...): plain text editors
- Word processors (Word, Writer, ...): WYSIWYG editors

WYSIWYG: What You See Is What You Get

A few years ago...

- No WYSIWYG editors
- Special marks for formatted text. For instance, old WordPerfect:
 - ***bold text***
 - _underlined text_

2. HTML

What is HTML?

- HTML stands for Hyper Text Markup Language
- HTML is the standard markup language for creating Web pages
- HTML describes the structure of a Web page
- HTML consists of a series of elements
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>Page Title</title>
  </head>
  <body>
    <h1>My First Heading</h1>
    <p>My first paragraph.</p>
  </body>
</html>
```

Example Explained

- The `<!DOCTYPE html>` declaration defines that this document is an HTML5 document
- The `<html>` element is the root element of an HTML page
- The `<head>` element contains meta information about the HTML page
- The `<title>` element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
- The `<body>` element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The `<h1>` element defines a large heading
- The `<p>` element defines a paragraph

2.1. HTML Elements

An HTML element tells the browser how to display the document. It is defined by a start tag, some content, and an end tag:

```
<tagname> Content goes here... </tagname>
```

Note: Tagnames are not case sensitive. The HTML standard does not require lowercase tags, but W3C **recommends** lowercase in HTML

The HTML **element** is everything from the start tag to the end tag:

```
<h1>My First Heading</h1>
```

```
<p>My first paragraph.</p>
```

Start tag	Element content	End tag
<h1>	My First Heading	</h1>
<p>	My first paragraph.	</p>
 	none	none

Note: Some HTML elements have no content (like the
 element). These elements are called empty elements. Empty elements do not have an end tag!

Nested HTML Elements

HTML elements can be nested (this means that elements can contain other elements). So, all HTML documents consist of nested HTML elements.

The example contains four HTML elements (<html>, <body>, <h1> and <p>):

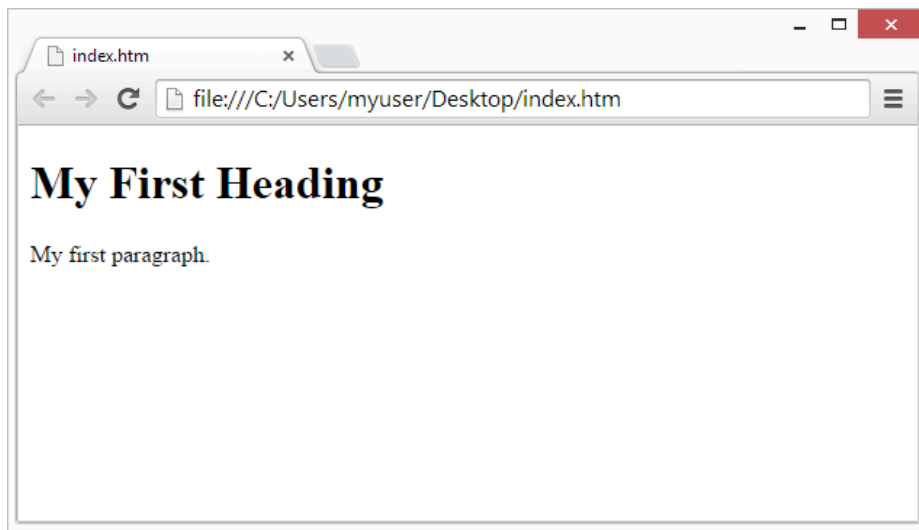
- The <html> element is the root element and it defines the whole HTML document.
- It has a start tag <html> and an end tag </html>.
- Then, inside the <html> element there is a <body> element:
 - The <body> element defines the document's body.
 - It has a start tag <body> and an end tag </body>.
 - Then, inside the <body> element there are two other elements: <h1> and <p>.

Some HTML elements will display correctly, even if you forget the end tag, but **never skip the end tag**.

2.2. Web Browsers

The purpose of a web browser (Chrome, Edge, Firefox, Safari) is to read HTML documents and display them correctly.

A browser does not display the HTML tags, but uses them to determine how to display the document:



View HTML Source Code:

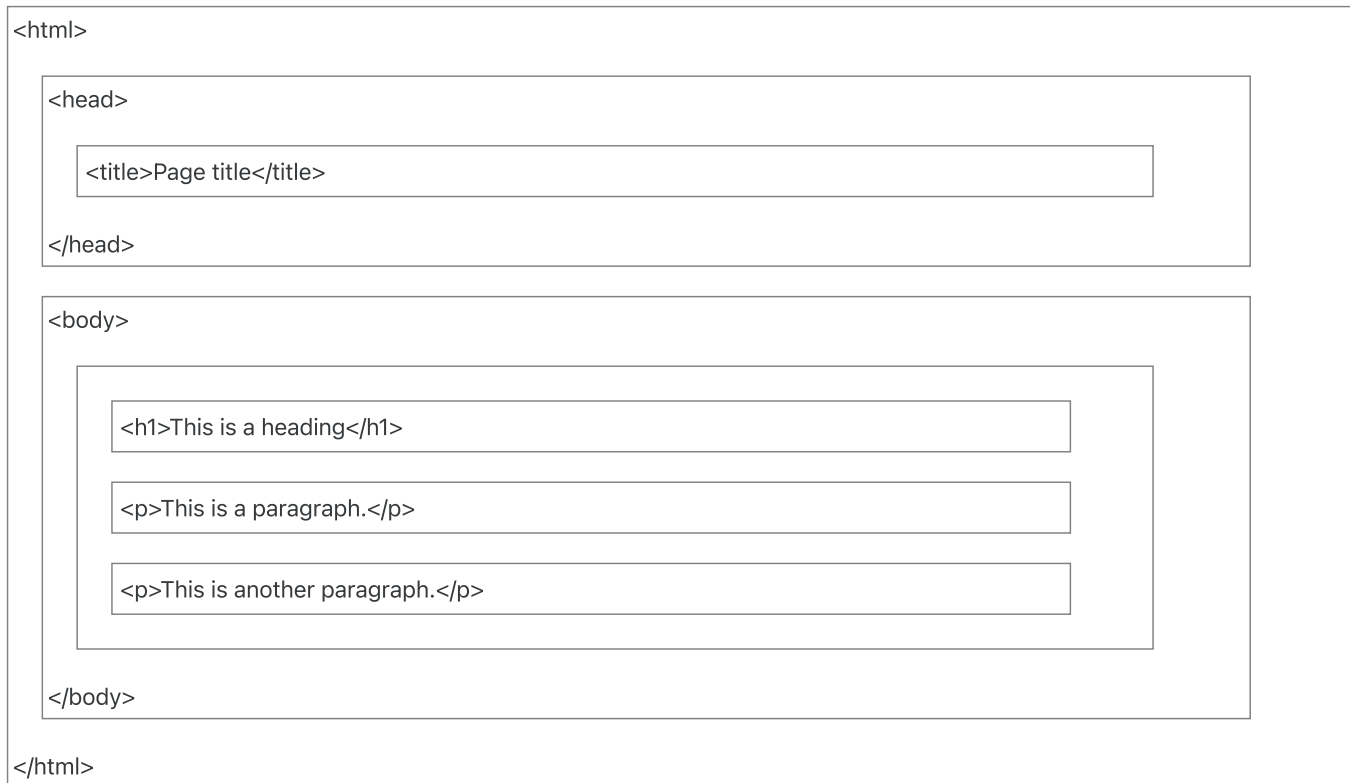
Right-click in an HTML page and select "View Page Source" (in Chrome) or "View Source" (in Edge), or similar in other browsers. This will open a window containing the HTML source code of the page.

Inspect an HTML Element:

Right-click on an element (or a blank area), and choose "Inspect" or "Inspect Element" to see what elements are made up of (you will see both the HTML and the CSS). You can also edit the HTML or CSS on-the-fly in the Elements or Styles panel that opens.

2.3. HTML Page Structure

Below is a visualization of an HTML page structure:



Note: The content inside the <body> section (the white area above) will be displayed in a browser. The content inside the <title> element will be shown in the browser's title bar or in the page's tab.

2.4. HTML Attributes

- All HTML elements can have **attributes**
- Attributes provide **additional information** about elements
- Attributes are always specified in **the start tag**
- Attributes usually come in name/value pairs like: **name="value"**
- Double quotes around attribute values are the most common in HTML, but single quotes can also be used depending of the situation.

Example:

```
<body bgcolor="blue">
```


3. Basic tags

HTML Paragraphs

HTML paragraphs are defined with the <p> tag:

Example

```
<p>This is a paragraph.</p>
```

```
<p>This is another paragraph.</p>
```

You cannot be sure how HTML will be displayed. Large or small screens, and resized windows will create different results. With HTML, you cannot change the display by adding extra spaces or extra lines in your HTML code.

The browser will automatically remove any extra spaces and lines when the page is displayed:

Example

```
<p>
```

```
This paragraph
```

```
contains a lot of lines
```

```
in the source code,
```

```
but the browser
```

```
ignores it.
```

```
</p>
```

```
<p>
```

```
This paragraph
```

```
contains      a lot of spaces
```

```
in the source      code,
```

```
but the      browser
```

```
ignores it.
```

```
</p>
```

HTML Headings

HTML headings are defined with the <h1> to <h6> tags.

<h1> defines the most important heading. <h6> defines the least important heading.

Example

```
<h1>Heading 1</h1>
```

```
<h2>Heading 2</h2>
```

```
<h3>Heading 3</h3>
```

```
<h4>Heading 4</h4>
```

```
<h5>Heading 5</h5>
```

```
<h6>Heading 6</h6>
```

HTML Comment Tag

You can add comments to your HTML source by using the following syntax:

```
<!-- Write your comments here -->
```

Notice that there is an exclamation point (!) in the start tag, but not in the end tag.

Example:

```
<!-- This is a comment -->
<p>This is a paragraph.</p>
<!-- Remember to add more information here -->
```

Other tags

- The `<br>` tag defines a line break.
- The `<hr>` tag draws a horizontal rule

4. Text formatting

Formatting elements were designed to display special types of text:

- `<b>` - Bold text
- `<strong>` - Important text
- `<i>` - Italic text
- `<em>` - Emphasized text
- `<mark>` - Marked text
- `<small>` - Smaller text
- `<del>` - Deleted text
- `<ins>` - Inserted text
- `<sub>` - Subscript text
- `<sup>` - Superscript text

The HTML `<strong>` element defines text with strong importance. The content inside is typically displayed in bold.

The HTML `<em>` element defines emphasized text. The content inside is typically displayed in italic.

5. Font

The tag specifies the font face, font size, and color of text.

The tag is not supported in HTML5. Use CSS instead.

Attribute	Value	Description
color	<i>rgb(x,x,x)</i>	Not supported in HTML5.
	<i>#xxxxxx</i>	Specifies the color of text
	<i>colorname</i>	
face	<i>font_family</i>	Not supported in HTML5. Specifies the font of text
size	<i>number</i>	Not supported in HTML5. Specifies the size of text

Examples

```
<font size="3" color="red">This is some text!</font>  
<font size="+3" color="yellow">This is some text!</font>  
<font size="2" color="blue">This is some text!</font>  
<font face="verdana" color="green">This is some text!</font>
```

6. Colors

HTML name	R	G	B
	Hex	Decimal	
Pink colors			
Pink	FF C0 CB	255	192 203
LightPink	FF B6 C1	255	182 193
HotPink	FF 69 B4	255	105 180
DeepPink	FF 14 93	255	20 147
PaleVioletRed	DB 70 93	219	112 147
MediumVioletRed	C7 15 85	199	21 133
Red colors			
LightSalmon	FF A0 7A	255	160 122
Salmon	FA 80 72	250	128 114
DarkSalmon	E9 96 7A	233	150 122
LightCoral	F0 80 80	240	128 128
IndianRed	CD 5C 5C	205	92 92
Crimson	DC 14 3C	220	20 60
FireBrick	B2 22 22	178	34 34
DarkRed	8B 00 00	139	0 0
Red	FF 00 00	255	0 0
Orange colors			
OrangeRed	FF 45 00	255	69 0
Tomato	FF 63 47	255	99 71
Coral	FF 7F 50	255	127 80
DarkOrange	FF 8C 00	255	140 0
Orange	FF A5 00	255	165 0
Yellow colors			
Yellow	FF FF 00	255	255 0
LightYellow	FF FF E0	255	255 224
LemonChiffon	FF FA CD	255	250 205
LightGoldenrodYellow	FA FA D2	250	250 210
PapayaWhip	FF EF D5	255	239 213
Moccasin	FF E4 B5	255	228 181
PeachPuff	FF DA B9	255	218 185
PaleGoldenrod	EE E8 AA	238	232 170
Khaki	F0 E6 8C	240	230 140
DarkKhaki	BD B7 6B	189	183 107
Gold	FF D7 00	255	215 0
Brown colors			
Cornsilk	FF F8 DC	255	248 220
BlanchedAlmond	FF EB CD	255	235 205
Bisque	FF E4 C4	255	228 196
NavajoWhite	FF DE AD	255	222 173
Wheat	F5 DE B3	245	222 179
BurlyWood	DE B8 87	222	184 135
Tan	D2 B4 8C	210	180 140
RosyBrown	BC 8F 8F	188	143 143
SandyBrown	F4 A4 60	244	164 96
Goldenrod	DA A5 20	218	165 32
DarkGoldenrod	B8 86 0B	184	134 11
Peru	CD 85 3F	205	133 63
Chocolate	D2 69 1E	210	105 30
SaddleBrown	8B 45 13	139	69 19
Sienna	A0 52 2D	160	82 45
Brown	A5 2A 2A	165	42 42
Maroon	80 00 00	128	0 0

HTML name	R	G	B			
	Hex	Decimal				
Green colors						
DarkOliveGreen	55	6B	2F	85	107	47
Olive	80	80	00	128	128	0
OliveDrab	6B	8E	23	107	142	35
YellowGreen	9A	CD	32	154	205	50
LimeGreen	32	CD	32	50	205	50
Lime	00	FF	00	0	255	0
LawnGreen	7C	FC	00	124	252	0
Chartreuse	7F	FF	00	127	255	0
GreenYellow	AD	FF	2F	173	255	47
SpringGreen	00	FF	7F	0	255	127
MediumSpringGreen	00	FA	9A	0	250	154
LightGreen	90	EE	90	144	238	144
PaleGreen	98	FB	98	152	251	152
DarkSeaGreen	8F	BC	8F	143	188	143
MediumAquamarine	66	CD	AA	102	205	170
MediumSeaGreen	3C	B3	71	60	179	113
SeaGreen	2E	8B	57	46	139	87
ForestGreen	22	8B	22	34	139	34
Green	00	80	00	0	128	0
DarkGreen	00	64	00	0	100	0
Cyan colors						
Aqua	00	FF	FF	0	255	255
Cyan	00	FF	FF	0	255	255
LightCyan	E0	FF	FF	224	255	255
PaleTurquoise	AF	EE	EE	175	238	238
Aquamarine	7F	FF	D4	127	255	212
Turquoise	40	E0	D0	64	224	208
MediumTurquoise	48	D1	CC	72	209	204
DarkTurquoise	00	CE	D1	0	206	209
LightSeaGreen	20	B2	AA	32	178	170
CadetBlue	5F	9E	A0	95	158	160
DarkCyan	00	8B	8B	0	139	139
Teal	00	80	80	0	128	128
Blue colors						
LightSteelBlue	B0	C4	DE	176	196	222
PowderBlue	B0	E0	E6	176	224	230
LightBlue	AD	D8	E6	173	216	230
SkyBlue	87	CE	EB	135	206	235
LightSkyBlue	87	CE	FA	135	206	250
DeepSkyBlue	00	BF	FF	0	191	255
DodgerBlue	1E	90	FF	30	144	255
CornflowerBlue	64	95	ED	100	149	237
SteelBlue	46	82	B4	70	130	180
RoyalBlue	41	69	E1	65	105	225
Blue	00	00	FF	0	0	255
MediumBlue	00	00	CD	0	0	205
DarkBlue	00	00	8B	0	0	139
Navy	00	00	80	0	0	128
MidnightBlue	19	19	70	25	25	111

HTML name	R	G	B
Hex	Decimal		
Purple, violet, and magenta colors			
Lavender	E6 E6 FA	230 230 250	
Thistle	D8 BF D8	216 191 216	
Plum	DD A0 DD	221 160 221	
Violet	EE 82 EE	238 130 238	
Orchid	DA 70 D6	218 112 214	
Fuchsia	FF 00 FF	255 0 255	
Magenta	FF 00 FF	255 0 255	
MediumOrchid	BA 55 D3	186 85 211	
MediumPurple	93 70 DB	147 112 219	
BlueViolet	8A 2B E2	138 43 226	
DarkViolet	94 00 D3	148 0 211	
DarkOrchid	99 32 CC	153 50 204	
DarkMagenta	8B 00 8B	139 0 139	
Purple	80 00 80	128 0 128	
Indigo	4B 00 82	75 0 130	
DarkSlateBlue	48 3D 8B	72 61 139	
SlateBlue	6A 5A CD	106 90 205	
MediumSlateBlue	7B 68 EE	123 104 238	

White colors			
White	FF	FF	FF
Snow	FF	FA	FA
Honeydew	F0	FF	F0
MintCream	F5	FF	FA
Azure	F0	FF	FF
AliceBlue	F0	F8	FF
GhostWhite	F8	F8	FF
WhiteSmoke	F5	F5	F5
Seashell	FF	F5	EE
Beige	F5	F5	DC
OldLace	FD	F5	E6
FloralWhite	FF	FA	F0
Ivory	FF	FF	F0
AntiqueWhite	FA	EB	D7
Linen	FA	F0	E6
LavenderBlush	FF	F0	F5
MistyRose	FF	E4	E1

Gray and black colors			
Gainsboro	DC	DC	DC
LightGray	D3	D3	D3
Silver	C0	C0	C0
DarkGray	A9	A9	A9
Gray	80	80	80
DimGray	69	69	69
LightSlateGray	77	88	99
SlateGray	70	80	90
DarkSlateGray	2F	4F	4F
Black	00	00	00

7. Images

The HTML `<img>` tag is used to embed an image in a web page.

Images are not technically inserted into a web page; images are linked to web pages. The `<img>` tag creates a holding space for the referenced image.

The `<img>` tag is empty, it contains attributes only, and does not have a closing tag.

The `<img>` tag has two required attributes:

- `src` - Specifies the path to the image
- `alt` - Specifies an alternate text for the image

Example:

```

```

Alternatively, you can use the `width` and `height` attributes:

```

```

If you have your images in a sub-folder, you must include the folder name in the `src` attribute:

```

```

To point to an image on another server, you must specify an absolute (full) URL in the `src` attribute:

```

```

8. Entities

Some characters are reserved in HTML. If you use the less than (<) or greater than (>) signs in your text, the browser might mix them with tags. Character entities are used to display reserved characters in HTML.

A character entity looks like this:

```
&entity_name;
```

OR

```
&#entity_number;
```

To display a less than sign (<) we must write: **<** or **<**;

Advantage of using an entity name: An entity name is easy to remember.

Disadvantage of using an entity name: Browsers may not support all entity names, but the support for entity numbers is good.

Useful HTML Character

Result	Description	Entity Name	Entity Number
	non-breaking space	 	
<	less than	<	<
>	greater than	>	>
&	ampersand	&	&
"	double quotation mark	"	"
'	single quotation mark (apostrophe)	'	'
¢	cent	¢	¢
£	pound	£	£
¥	yen	¥	¥
€	euro	€	€
©	copyright	©	©
®	registered trademark	®	®

Diacritical Marks

Lowercase Letters

Char	ALT Code	HTML	MAC	Description
For MAC, ... means release keys before continuing				
á	ALT + 0225	á	OPTION + E ... A	lowercase a with acute accent
â	ALT + 0226	â	OPTION + I ... A	lowercase a with circumflex
à	ALT + 0224	à	OPTION + ` ... A	lowercase a with grave accent
å	ALT + 0229	å	OPTION + SHIFT + A	lowercase a with ring / Nordic / Angstrom
ã	ALT + 0227	ã	OPTION + N ... A	lowercase a with tilde
ä	ALT + 0228	ä	OPTION + U ... A	lowercase a with umlaut/diaeresis
æ	ALT + 0230	æ	OPTION + ' (single quote)	lowercase ae ligature
ç	ALT + 0231	ç	OPTION + C	lowercase c with cedilla
é	ALT + 0233	é	OPTION + E ... E	lowercase e with acute accent
ê	ALT + 0234	ê	OPTION + I ... E	lowercase e with circumflex
è	ALT + 0232	è	OPTION + ` ... E	lowercase e with grave accent
ë	ALT + 0235	ë	OPTION + U ... E	lowercase e with umlaut/diaeresis
í	ALT + 0237	í	OPTION + E ... I	lowercase i with acute accent
î	ALT + 0238	î	OPTION + I ... I	lowercase i with circumflex
ì	ALT + 0236	ì		lowercase i with grave accent
ï	ALT + 0239	ï	OPTION + U ... I	lowercase i with umlaut/diaeresis

Char	ALT Code	HTML	MAC	Description
ñ	ALT + 0241	ñ	OPTION + N ... N	lowercase n with tilde
ó	ALT + 0243	ó	OPTION + E ... O	lowercase o with acute accent
ô	ALT + 0244	ô	OPTION + I ... O	lowercase o with circumflex
ò	ALT + 0242	ò		lowercase o with grave accent
ø	ALT + 0248	ø	OPTION + O	lowercase o with slash
õ	ALT + 0245	õ	OPTION + N ... O	lowercase o with tilde
ö	ALT + 0246	ö	OPTION + U ... O	lowercase o with umlaut/diaeresis
ő	ALT + 0337	ő		lowercase o with double acute accent
œ	ALT + 0156	œ	OPTION + Q	lowercase oe ligature
ú	ALT + 0250	ú	OPTION + E ... U	lowercase u with acute accent
û	ALT + 0251	û	OPTION + I ... U	lowercase u with circumflex
ù	ALT + 0249	ù	OPTION + ` ... U	lowercase u with grave accent
ü	ALT + 0252	ü	OPTION + U ... U	lowercase u with umlaut/diaeresis
ű	ALT + 0369	ű		lowercase u with double acute accent
ý	ALT + 0253	ý		lowercase y with double acute accent
ÿ	ALT + 0255	ÿ	OPTION + U ... Y	lowercase y with umlaut/diaeresis

Uppercase Letters

Char	ALT Code	HTML	MAC	Description
For MAC, ... means release keys before continuing				
Á	ALT + 0193	Á	OPTION + E ... SHIFT + A	uppercase A with acute accent
Â	ALT + 0194	Â	OPTION + I ... SHIFT + A	uppercase A with circumflex
À	ALT + 0192	À	OPTION + ` ... SHIFT + A	uppercase A with grave accent
Å	ALT + 0197	Å	OPTION + SHIFT + A	"A" with ring / Nordic / Angstrom
Ã	ALT + 0195	Ã	OPTION + N ... SHIFT + A	uppercase A with tilde
Ä	ALT + 0196	Ä	OPTION + U ... SHIFT + A	uppercase A with umlaut/diaeresis
Æ	ALT + 0198	Æ	OPTION + SHIFT + ' (single quote)	uppercase AE ligature
Ç	ALT + 0199	Ç	OPTION + SHIFT + C	uppercase C with cedilla
É	ALT + 0201	É	OPTION + E ... SHIFT + E	uppercase E with acute accent
Ê	ALT + 0202	Ê	OPTION + I ... SHIFT + E	uppercase E with circumflex
È	ALT + 0200	È	OPTION + ` ... SHIFT + E	uppercase E with grave accent
Ë	ALT + 0203	Ë	OPTION + U ... SHIFT + E	uppercase E with umlaut/diaeresis
Í	ALT + 0205	Í	OPTION + E ... SHIFT + I	uppercase I with acute accent
Î	ALT + 0206	Î	OPTION + I ... SHIFT + I	uppercase I with circumflex
Ì	ALT + 0204	Ì		uppercase I with grave accent
Ï	ALT + 0207	Ï	OPTION + U ... SHIFT + I	uppercase I with umlaut/diaeresis
Ñ	ALT + 0209	Ñ	OPTION + N ... SHIFT + N	uppercase N with tilde
Ó	ALT + 0211	Ó	OPTION + E ... SHIFT + O	uppercase O with acute accent
Ô	ALT + 0212	Ô	OPTION + I ... SHIFT + O	uppercase O with circumflex
Ø	ALT + 0216	Ø	OPTION + SHIFT + O	uppercase O with slash
Õ	ALT + 0213	Õ	OPTION + N ... SHIFT + O	uppercase O with tilde
Ö	ALT + 0214	Ö	OPTION + U ... SHIFT + O	uppercase O with umlaut/diaeresis
Ő	ALT + 0336	Ő		uppercase O with double acute accent
Œ	ALT + 0140	Œ	OPTION + SHIFT + Q	uppercase OE ligature
ß	ALT + 0223	ß	OPTION + S	eszett (sz), sharp / double "s"
Ú	ALT + 0218	Ú	OPTION + E ... SHIFT + U	uppercase U with acute accent
Û	ALT + 0219	Û	OPTION + I ... SHIFT + U	uppercase U with circumflex
Ù	ALT + 0217	Ù	OPTION + ` ... SHIFT + U	uppercase U with grave accent
Ü	ALT + 0220	Ü	OPTION + U ... SHIFT + U	uppercase U with umlaut/diaeresis
Ű	ALT + 0368	Ű		uppercase U with double acute accent
Ý	ALT + 0221	Ý		uppercase Y with acute accent
Ÿ	ALT + 0159	Ÿ	OPTION + U ... SHIFT + Y	uppercase Y with umlaut/diaeresis

9. Atribut align

The purpose of the HTML align attribute is to specify the alignment of data and the justification of text in a cell of a table.

Supported Tags:

- <applet>
- <caption>
- <col>
- <colgroup>
- <hr>
- <iframe>
-
- <p>
- <table>
- <tbody>
- <td>
- <tfoot>
- <th>
- <thead>
- <tr>

Syntax:

```
<element_name align="left | right | center | justify">
```

Attribute Values:

- left: It sets the text left-align.
- right: It sets the text right-align.
- center: It sets the text center-align.
- justify: It stretch the text of paragraph to set the width of all lines equal.

Examples:

```
<p align="left">Left align content</p>
```

```
<p align="center">center align content</p>
```

```
<p align="right">Right align content</p>
```

10. Links

Text or graphic area that takes us to another document or to another position within the same document. The linked document does not need to be "nearby" (Internet).

We will use the **A** (anchor) tag.

```
<A HREF="address">Link text</A>
```

The address must be in the URL (Uniform Resource Locator) format:

```
service://machine:port/path/file#section
```

Example:

```
http://www.xtec.cat
```

Link types

- **Absolute links:** specify a unique document on the Internet.

```
service://machine:port/route/file
```

- **Relative links:** specify a document positioned relative to the current document.

```
path/file
```

- **Internal links:** specify a part of the document (anchors).

```
file#section
```

Services

- http
- https: http service with built-in security (encryption, etc.).
- ftp
- mailto
- news
- telnet

Relative links

They specify a document positioned relative to the current document.

- Different folders (directories) are separated with /
- To indicate an upper folder, use ..

Examples:

```
<A HREF="courses/index.html">Link</A>
```

```
<A HREF="../index.html">Link</A>
```

Internal links

With internal links it is possible to access different positions in a document. The position you want to access must be specified:

```
<A NAME="some_name">
```

To link from the same document:

```
<A HREF="#some_name">Section Access</A>
```

To link to it from another document:

```
<A HREF="file.html#some_name">Section access</A>
```

11. Lists

Lists are a resource we use to make an enumeration. There are three types of list:

- ordered lists.
- unordered lists.
- definition lists.

Although initially the ordered `<ol>` and disordered `<ul>` lists are very differentiated, from CSS we can modify the aesthetics and get the same look.

Ordered lists

Ordered lists are the lists in which we need the items to be numbered. To mark the beginning and end of the list we use the tag `<ol>` (ordered list) and to mark each item in the list we use the tag `<li>` (list item).

Imagine we are making a recipe and want to list the steps we need to follow. The code would be as follows:

```
<h2> Preparation </h2>
<ol>
  <li> Beat the eggs </li>
  <li> Heat the oil in a frying pan </li>
  <li> Throw the beaten eggs in the pan </li>
  <li> Wrap the omelette </li>
</ol>
```

Unordered lists

Unordered lists are lists in which we do not need items to be numbered, that is, the order of the enumeration is not required. To mark the beginning and end of the list we use the tag `<ul>` (unordered list) and to mark each item in the list we use the tag `<li>` (list item).

If, for example, we want to list all the ingredients needed to make a French tortilla, we code it as follows:

```
<h2> Ingredients </h2>
<ul>
  <li> Eggs </li>
  <li> Sal </li>
  <li> Oil </li>
</ul>
```

Definition lists

Definition lists are characterized by being an enumeration of definitions of terms. Therefore, the items in the definition lists are made up of two elements: the term and the definition of that term. To mark the beginning and end of the list we use the tag `<dl>` (definition list); the term `<dt>` (definition term) is used for each of the terms, and the `<dd>` (definition description) tag is used for each of the definitions.

For example, if we want to define the terms HTML, CSS and JavaScript, we can use this type of list.

```
<dl>
  <dt> HTML </dt>
  <dd> Markup language designed to structure text. </dd>
  <dt> CSS </dt>
  <dd> Language created to define the appearance of documents. </dd>
  <dt> JavaScript </dt>
  <dd> Interpreted programming language. </dd>
</dl>
```

Nested lists

If we want to make complex lists, that is, insert lists into other lists, we can do the same. All we have to do is open and close the tags correctly: if we want to encapsulate a list, either numbered or unnumbered. This list must be in an item (`<li>`) in the parent list. As an example, we can think of steps for making a web design of the client environment:

```

<h1> Phases in web design. </h1>
<ol>
  <li> Sketch. </li>
  <li> Making the prototype. </li>
  <ol>
    <li> Choose design tools. </li>
    <ul>
      <li> Typographic authoring tool </li>
      <li> Color Scheme Creation Tool </li>
      <li> Drawing tool </li>
    </ul>
    <li> Creating the style guide. </li>
  </ol>
  <li> Layout. </li>
  <ul>
    <li> Choice of languages. </li>
    <li> Testing in different browsers. </li>
    <li> Tests at different resolutions. </li>
  </ul>
  <li> Delivery of templates to the customer. </li>
</ol>

```

Other

Other characteristics can be defined through attributes such as peak type, numbering, etc.

```

<p> Unordered list with type attribute: </p>
<ul>
  <li type = "disc"> First element </li>
  <li type = "square"> Second element </li>
  <li type = "circle"> Third element </li>
</ul>

```

```

<p> Ordered list with start attribute: </p>
<ol start = 12>
  <li> First item </li>
  <li> Second item in the list. </li>
  <li> Third item, etc. </li>
</ol>

<!-- type: a, A, i o I -->
<p> Ordered list with type attribute: </p>
<ol type = "a">
  <li> First item </li>
  <li> Second item in the list. </li>
  <ol>
    <li> First sublist item </li>
    <li> Second sublist item </li>
    <li> Third item sublists </li>
  </ol>
</ol>

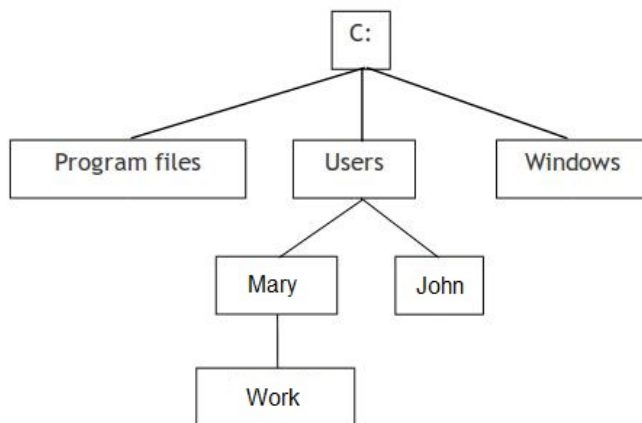
```

12. Paths

- **Files** are storage units from the user's point of view.
 - Generally, files are identified by their name and their extension, although sometimes the extension is not necessary, or simply it hides.
- **Folders** are containers that can hold more than one at a time folders or files.
- The graphical way to represent folders and subfolders of a disk is the **tree structure**, as it represents a tree inverted, or if you want, the part of the roots.
 - The idea is that each folder can contain folders within it inside we call them subfolders), and so recursively.

Folder structure

An example of folder structure in Windows could be:



Paths

- The top node of the folder tree structure is called the **root**. In Windows it is the name of a drive (C:, D:, etc.). In Linux operating systems the root is `/`.
 - The path from the root to a specific folder or file is called **absolute path**. Example: C:\Windows, C:\Users\Mary\Work, ...
 - The path from where you are to a particular folder or file is called **relative path**. We must take into account that `..` indicates the upper folder (immediately above). Example (we are in the Users folder): Maria\Work, ../Windows, ...

Web pages

In web pages design the relative path is always used, since it allows to move an entire website without any problem. It should be noted that some programs that allow you to create web pages, depending on the action taken, use the absolute path. You need to correct it by hand.

13. Tables

HTML tables allow web developers to arrange data into rows and columns. A table in HTML consists of table cells inside rows and columns.

Example:

```
<table>
  <tr>
    <td>Emil</td>
    <td>Tobias</td>
    <td>Linus</td>
  </tr>
</table>
```

Each **table cell** is defined by a **<td>** and a **</td>** tag. td stands for table data. Everything between <td> and </td> are the content of the table cell.

A table cell can contain all sorts of HTML elements: text, images, lists, links, other tables, etc.

Each **table row** starts with a **<tr>** and ends with a **</tr>** tag. tr stands for table row. You can have as many rows as you like in a table; just make sure that the number of cells are the same in each row.

Table Headers

Sometimes you want your cells to be **table header cells**. In those cases use the **<th>** tag instead of the <td> tag:

```
<table>
  <tr>
    <th>Name</th>
    <th>Surname</th>
    <th>Alias</th>
  </tr>
  <tr>
    <td>Emil</td>
    <td>Tobias</td>
    <td>Linus</td>
  </tr>
</table>
```

th stands for table header.

To use the first column as table headers, define the first cell in each row as a <th> element:

```
<table>
  <tr>
    <th>Firstname</th>
    <td>Jill</td>
    <td>Eve</td>
  </tr>
  <tr>
    <th>Lastname</th>
    <td>Smith</td>
    <td>Jackson</td>
  </tr>
  <tr>
    <th>Age</th>
    <td>94</td>
    <td>50</td>
  </tr>
</table>
```

Attributes

Some attributes:

- border

- width, height
- bgcolor
- align: left, center, right
- valign: top, middle, bottom

Example:

```
<table width=50% bgcolor="yellow" border=1>
  <caption align="bottom">Example of a table</caption>
  <tr bgcolor="green">
    <th>Name</th>
    <th>Surname</th>
    <th>Alias</th>
  </tr>
  <tr>
    <td bgcolor="red"><font color="white"><i>Emil</i></font></td>
    <td width=500 height=50 align="center" valign="top">Tobias</td>
    <td>Linus</td>
  </tr>
</table>
```

Padding & Spacing

HTML tables can adjust the padding inside the cells, and also the space between the cells.



Example:

```
<table cellpadding=10 cellspacing=5>
```

Colspan & Rowspan

HTML tables can have cells that span over multiple rows and/or columns.



To make a cell span over multiple columns, use the colspan attribute:

```
<table>
  <tr>
    <th colspan="2">Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Jill</td>
    <td>Smith</td>
    <td>43</td>
  </tr>
  <tr>
    <td>Eve</td>
    <td>Jackson</td>
    <td>57</td>
  </tr>
</table>
```

To make a cell span over multiple rows, use the rowspan attribute:

```
<table>
  <tr>
    <th>Name</th>
    <td>Jill</td>
  </tr>
  <tr>
    <th rowspan="2">Phone</th>
    <td>555-1234</td>
  </tr>
  <tr>
    <td>555-8745</td>
  </tr>
</table>
```

14. Frames

Definition

- Documents without content.
- Function of splitting the screen into multiple areas.
- The **content** of a frame is an **HTML file**.
- Frames do not have **BODY**, they have the **FRAMESET** tag instead.

Structure

```
<html>
  <frameset ...>
    More specifications about frames
  </frameset>
  <noframes>
    <P>This page can only be viewed if the browser supports frames.</P>
  </noframes>
</html>
```

Size

The size of the frame can be indicated with:

- Percentage
- Absolute value
- About the rest

Example:

```
<FRAMESET COLS="10%,*,200,2*">
<FRAMESET ROWS="10%,*,200,2*">
```

Nesting

```
<FRAMESET COLS="20%,80%">
  <FRAME NAME="index" SRC="index.html">
  <FRAMESET ROWS="*,80">
    <FRAME NAME="menu" SRC="me.html">
    <FRAME NAME="example" SRC="ex.html">
  </FRAMESET>
</FRAMESET>
```

Attributes

- name
- src
- scrolling="yes" | "no" | "auto": controls the appearance of scrollbars.
- noresize: prevents resizing frames.
- frameborder=number: indicates the size of the border.

Links

The hyperlinks of each of the frames can be a link for the same frame, the entire document or in another frame.

```
<a href=address target="name">
```

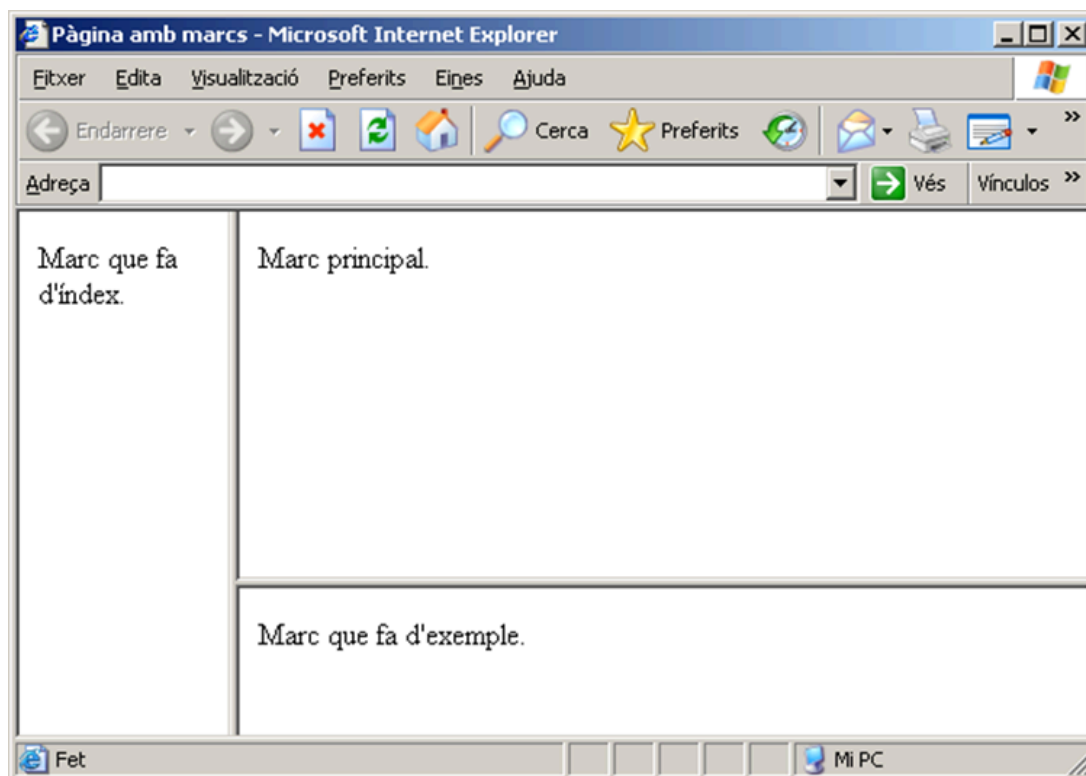
where name is the name given to the corresponding frame.

Special redirections:

- _blank: exit the frameset and go to a new window.
- _self: loads the new page in the same frameset (default).
- _relative: load the new page in the immediate top frameset.
- _top: loads the new page in the absolute top frameset.

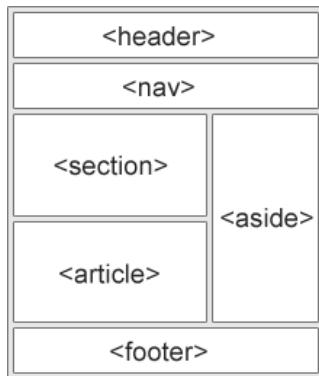
Example

```
<FRAMESET COLS="20%,80%">
  <FRAME NAME="index" SRC="index.html">
  <FRAMESET ROWS="*,80">
    <FRAME NAME="menu" SRC="me.html">
    <FRAME NAME="example" SRC="ex.html">
  </FRAMESET>
</FRAMESET>
```



15. Layout

Page layout is the part of graphic design that deals with the **arrangement** of visual elements on a page. Page layout is used to make the web pages look better. It establishes the overall **appearance**, relative importance, and relationships between the graphic elements to achieve a smooth flow of information and eye movement for maximum effectiveness or impact.



This design can be done using three techniques:

- Using tables
- Using frames
- Using div or span (layers): widely used in CSS

16. Bibliography

- <https://www.w3schools.com/html>